

Derivatives of Inverse and Inverse-Trigonometric Functions

Answer Key

AP Calculus AB/BC — Differentiation

Quick Answers

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|---|---|
| 1. $\frac{3}{9x^2+1}$ | 6. $-\frac{2x}{(x^2+1)^2}$ |
| 2. $\frac{2x}{\sqrt{1-x^4}}$ | 7. $\frac{1}{3}$ |
| 3. $1, \frac{1}{5}$ | 8. $\left(\operatorname{atan}(x) + \frac{1}{x^2+1}\right)e^x$ |
| 4. $\frac{x}{x^2+1} + \operatorname{atan}(x)$ | 9. $\frac{1}{2\sqrt{x(x+1)}}$ |
| 5. $-\frac{2}{\sqrt{1-4x^2}}$ | 10. $-\frac{16x}{(4x^2+1)^2}$ |
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Curriculum

Level: AP Calculus AB/BC — Differentiation

FUN-3.D–FUN-3.E – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 11 tagged checks.

Every solution names the inner function u (or the rule being combined) before it differentiates. On the AP exam the chain-rule factor u' is the mark most often lost, so it is written explicitly in each step.

1. Differentiate $y = \arctan(3x)$.

Solution: $u = 3x$, so $u' = 3$, and $\frac{d}{dx} \arctan(u) = \frac{u'}{1+u^2}$:

$$y' = \frac{3}{1+(3x)^2}$$

$$y' = \frac{3}{1+9x^2}$$

2. Differentiate $y = \arcsin(x^2)$.

Solution: $u = x^2$, so $u' = 2x$, and $\frac{d}{dx} \arcsin(u) = \frac{u'}{\sqrt{1-u^2}}$:

$$y' = \frac{2x}{\sqrt{1-(x^2)^2}}$$

$$y' = \frac{2x}{\sqrt{1-x^4}}$$

Common error: writing $\sqrt{1-x^2}$ in the denominator — the square is applied to u , not to x .

3. $f(x) = x^3 + 2x + 1$; find b with $f(b) = 4$, then $(f^{-1})'(4)$.

(a) **Solution:** Solve $x^3 + 2x + 1 = 4$, i.e. $x^3 + 2x - 3 = 0$. Factoring gives $(x-1)(x^2+x+3) = 0$, and the quadratic has no real root, so the only real solution is $b = 1$

(b) **Solution:** $f'(x) = 3x^2 + 2$, so $f'(1) = 5$. The inverse-function rule $(f^{-1})'(a) = \frac{1}{f'(b)}$

with $b = 1$ gives

$$(f^{-1})'(4) = \frac{1}{5}$$

Note: the reciprocal is of f' evaluated at the *input* b , never at the output a ; $1/f'(4) = 1/50$ is the classic wrong answer.

4. Differentiate $y = x \arctan(x)$.

Solution: Product rule with $f = x$, $g = \arctan x$:

$$y' = (1) \arctan(x) + x \cdot \frac{1}{1+x^2}$$

$$y' = \arctan(x) + \frac{x}{1+x^2}$$

5. Differentiate $y = \arccos(2x)$.

Solution: $u = 2x$, $u' = 2$, and $\frac{d}{dx} \arccos(u) = \frac{-u'}{\sqrt{1-u^2}}$ — the arccos derivative is the negative of the arcsin one:

$$y' = \frac{-2}{\sqrt{1-(2x)^2}}$$

$$y' = \frac{-2}{\sqrt{1-4x^2}}$$

6. Second derivative of $y = \arctan(x)$.

Solution: $y' = \frac{1}{1+x^2} = (1+x^2)^{-1}$. Differentiate again with the chain rule (inner $1+x^2$, derivative $2x$):

$$y'' = -1(1+x^2)^{-2}(2x)$$

$$y'' = \frac{-2x}{(1+x^2)^2}$$

Reading: y'' is positive for $x < 0$ and negative for $x > 0$, matching the S-shape of $\arctan$ with its inflection at the origin.

7. $g(x) = 2x + \sin(x)$; compute $(g^{-1})'(0)$.

Solution: First the input: $g(0) = 0 + \sin 0 = 0$, so $b = 0$. Then $g'(x) = 2 + \cos(x)$, giving $g'(0) = 2 + 1 = 3$, and $(g^{-1})'(0) = \frac{1}{3}$

Since $g'(x) = 2 + \cos x \geq 1 > 0$, g is strictly increasing everywhere, which is what guarantees the inverse exists.

8. Differentiate $y = e^x \arctan(x)$.

Solution: Product rule; the derivative of e^x is itself.

$$y' = e^x \arctan(x) + e^x \cdot \frac{1}{1+x^2}$$

$$y' = e^x \left(\arctan(x) + \frac{1}{1+x^2} \right)$$

9. Differentiate $y = \arctan(\sqrt{x})$.

Solution: $u = \sqrt{x} = x^{1/2}$, so $u' = \frac{1}{2\sqrt{x}}$ and $u^2 = x$:

$$y' = \frac{1}{1+x} \cdot \frac{1}{2\sqrt{x}}$$

$$y' = \frac{1}{2\sqrt{x}(1+x)}$$

Why the denominator collapses: $(\sqrt{x})^2 = x$ exactly, so no radical survives in the first factor.

10. Challenge: second derivative of $y = \arctan(2x)$.

Solution: First derivative by the chain rule ($u = 2x$, $u' = 2$): $y' = \frac{2}{1+4x^2} = 2(1+4x^2)^{-1}$. Differentiate again, chain rule on $1+4x^2$ (derivative $8x$):

$$y'' = 2 \cdot (-1)(1+4x^2)^{-2}(8x)$$

$$y'' = \frac{-16x}{(1+4x^2)^2}$$

Check the structure: y'' vanishes only at $x = 0$, the inflection point of $\arctan(2x)$, and the outer constant 2 times the inner $8x$ is where the 16 comes from.